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Paleomagnetism and Tectonic Interpretations of the Taos Plateau Volcanic Field, Rio Grande Rift, New Mexico

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The tectonic response of the Taos Plateau volcanic field in the southern San Luis basin to the late stage extensional environment of the Rio Grande rift was investigated using paleomagnetic techniques. Sixty-two sites (533 samples) of Pliocene volcanic units were collected covering four major rock types with ages of 4.7 to 1.8 Ma. Twenty-two of these sites were from stratigraphic sections of the lower, middle and upper Servilleta Basalt collected in the Rio Grande gorge at two locations 19 km apart. Flows from the lower and middle members in the southern gorge record reversed polarities, while those in Garapata Canyon are normal with an excursion event in the middle of the sequence. The uppermost flows of the upper member at both sites display normal directions. Although these sections correlate chemically, they seem to represent different magnetic time periods during the Gilbert Reversed-Polarity Chron. Alternating field demagnetization, aided by principal component analysis, yields 55 sites with stable directions representing both normal and reversed polarities, and five sites indicating transitional fields. Mean direction of the normal and inverted reversed sites is I=49.3° and D=356.7° (a_{95}=3.6°). Angular dispersion of the virtual geomagnetic poles is 16.3°, which is consistent with paleosecular variation model G, fit to data from the past 5 m.y. Comparison with the expected direction indicates no azimuthal rotation of the Taos Plateau volcanic field; inclination flattening for the southern part of the plateau is 8.3° ± 5.3°. Previous paleomagnetic data indicate 10°-15° counterclockwise rotation of the Espanola block to the south over the past 5 m.y. The data suggest the Taos Plateau volcanic field, showing no rotation and some flattening in the south and east, has acted as a stable buttress and has been downwarped by overriding of the southeastern end of the plateau by the Picuris Mountains, which make up the northern corner of the counterclockwise rotating Espanola block.

INTRODUCTION

Previous structural and paleomagnetic work has identified a large diamond-shaped block in the north central Rio Grande rift that has undergone significant counterclockwise rotation [*Muehlberger*, 1979; *Dungan et al.*, 1984; *Brown and Golombek*, 1985; 1986; *Aldrich*, 1986]. The block extends from the San Luis basin in the north to the Albuquerque basin in the south and includes the eastern part of the intervening Espanola basin (Figure 1). It is bounded by the following recently active major fault zones: the Embudo fault to the north [*Muehlberger*, 1979; *Dungan et al.*, 1984; *Aldrich and Dethier*, 1990], the Pajarito fault zone to the west [*Golombek*, 1983], the Picuris-Pecos fault to the east [*Montgomery*, 1953; *Miller et al.*, 1963] and the Tijeras-Canoncito fault zone to the south [*Lisenbee et al.*, 1979]. Extensive structural and paleomagnetic work on the timing and structural consequences of counterclockwise block rotation (discussed more fully by *Brown and Golombek*, [1986] supports motion beginning at about 5 Ma that resulted in uplift of the acute corners (Picuris and Sandia uplifts) and subsidence of the obtuse corners (Velarde graben and Santa Fe embayment). Counterclockwise rotation of crustal blocks could be driven by left slip along the rift, which has been suggested previously [*Kelley*, 1979; 1982; *Muehlberger*, 1979; *Dungan et al.*, 1984].

The Taos Plateau volcanic field (TPVF) lies within the southern portion of the San Luis basin in the northern Rio Grande rift (Figure 1), north of the proposed rotated block. The San Luis basin is the northernmost of three north-trending, right-echelon, asymmetric, fault-bounded basins that make up the central Rio Grande rift. The basin is bounded on the east and the west by the Precambrian-cored Sangre de Cristo and Tusas mountains, respectively. It is separated from the Espanola basin on the south by a Precambrian basement high [*Manley*, 1979] and by the Embudo fault along which the Picuris Mountains have been uplifted in the past 5 m.y. [*Muehlberger*, 1979; *Dungan et al.*, 1984]. Within the southern San Luis basin is the Taos graben, along which most of the recent activity has been concentrated [*Lipman and Mehnert*, 1979]. It is bounded on the east by the range-bounding fault at the front of the Sangre de Cristo Mountains (estimated to have had 7-8 km of displacement) and on the west by an east-dipping normal fault inferred to be beneath the Rio Grande. A horst abuts the Taos graben on the west exposing early Miocene intrusives and intermediate composition volcanic and pyroclastic rocks attributed to prerift (prior to 26 Ma) [*Hagstrum and Lipman*, 1986] and early rift phases of volcanism [*Lipman and Mehnert*, 1979; *Dungan et al.*, 1984; *Thompson et al.*, 1986].

The TPVF (Figure 2) includes the Servilleta Basalt (olivine tholeiite) and lesser amounts of basaltic andesite, andesite, dacite and rhyolite that were erupted from about 5 to 1.5 Ma [*Lipman and Mehnert*, 1979; *Dungan et al.*, 1984]. A unified magmatic system is suggested by the spatial relationship to basaltic shields on the west margin of the TPVF surrounded, in turn, by andesite and dacite [*Lipman and Mehnert*, 1979]; (Figure 2). The lower, middle, and upper Servilleta basalts (LSBs, MSBs and USBs) have been recognized on the basis of field relationships and chemical characteristics and were deposited in short pulses separated by long hiatuses [*Dungan et al.*, 1984; 1986].

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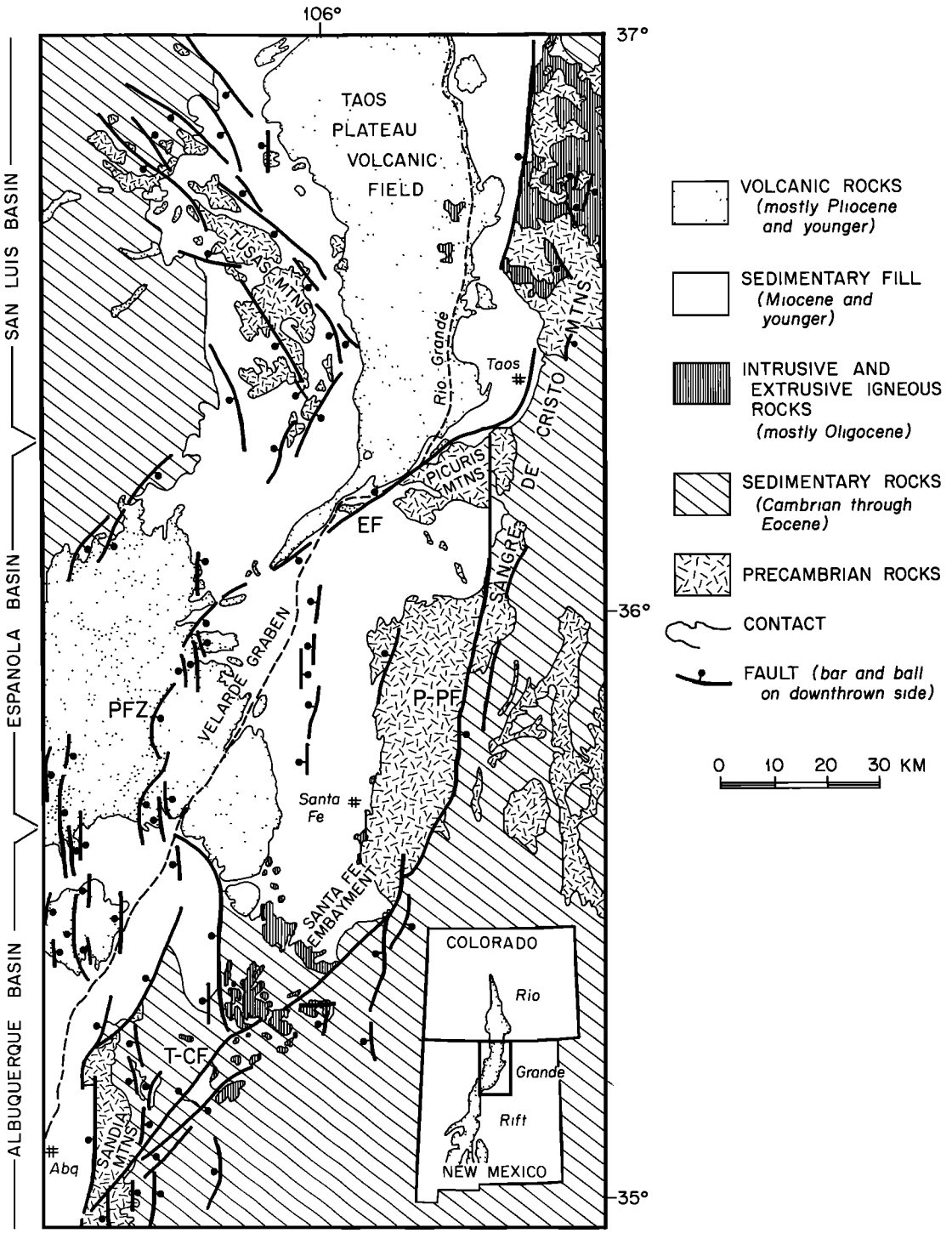


Fig. 1. Generalized geologic map of the northern Rio Grande rift, New Mexico. The Taos Plateau volcanic field lies within the southern San Luis Basin. Some important faults within the rift are the Embudo fault, EF; the Pajarito fault zone, PFZ; the Picuris-Pecos fault, P-P and the Tijeras-Canoncito fault zone, T-CF.

In this paper we report on paleomagnetic data from the TPVF in order to answer the following questions. Has the TPVF undergone rotation, and if not, has it been a stable buttress? Has the TPVF undergone warping from the compression and uplift of the Picuris Mountains? In addition, the stratigraphic and geochemical work of *Lipman and Mehnert* [1979] and *Dungan et al.* [1984; 1986] in the TPVF makes it possible to investigate correlations between rock type, age, and magnetic polarity directions and to integrate these with magnetic reversals and K-Ar ages.

PALEOMAGNETIC TECHNIQUES AND RESULTS

Sixty-two sites were collected to sample the full spatial, temporal, and compositional diversity of the TPVF (Figure 2), with each site representing one unique lava flow. Forty sites on the surface of the plateau were sampled, where possible at sites with published K-Ar age dates [*Lipman and Mehnert*, 1979; *Baldridge et al.*, 1980; *Ozima et al.*, 1967]. Twelve flows in the southern

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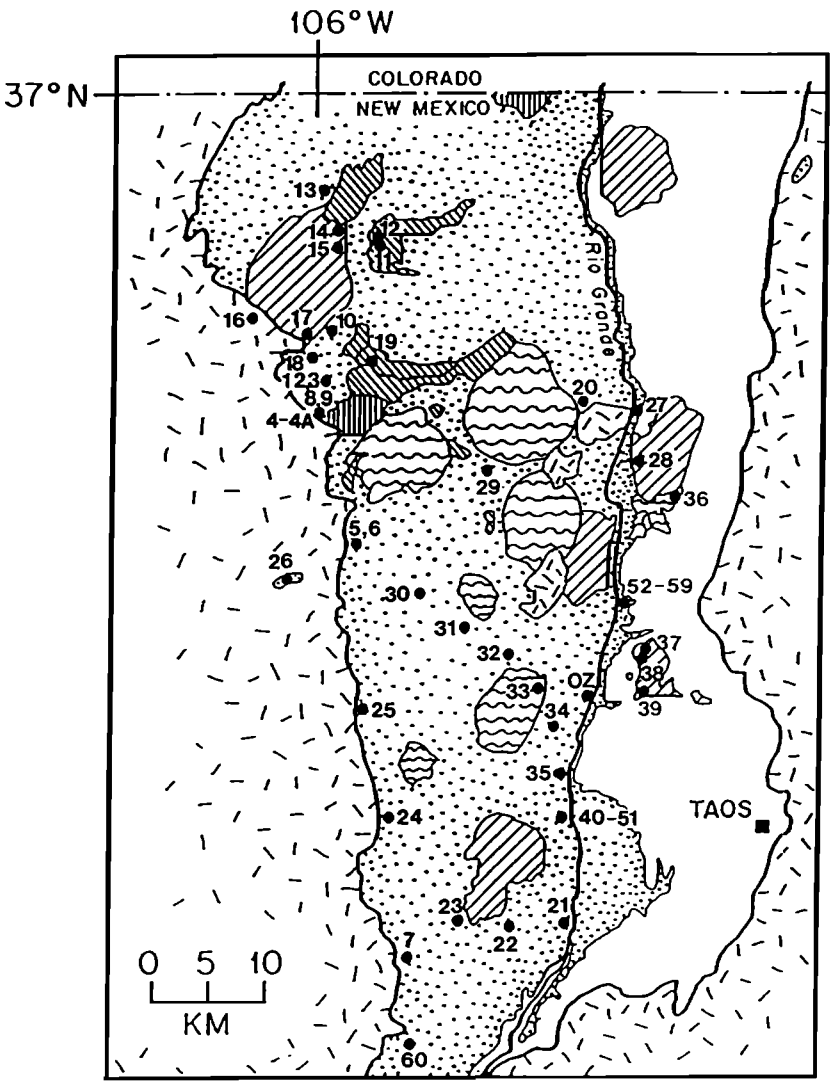


Fig. 2. Generalized geologic map of the Taos Plateau volcanic field, northern New Mexico. Numbered circles refer to paleomagnetic sites listed in Table 1. Patterns: dashed, Miocene and earlier undifferentiated rocks; dotted, Servilleta Basalt; wavy, andesite; vertical lined, rhyolite; NE-SW lined, dacite; NW-SE lined, xenocrystic basaltic andesite; plain, Quaternary basin fill. OZ refers to Dunn Bridge location of *Ozima et al.* [1967]. Figure is adapted from *Lipman and Mehnert* [1979].

Rio Grande gorge (sites 40-51) and 10 flows in Garapata Canyon (sites 52-59) were collected in stratigraphic sequence; these correspond to sections 4 and 10 of the Servilleta Basalt sequences of *Dungan et al.* [1984]. Five to 14 samples (average eight) were taken at each flow using a gas-powered rock drill [*Doell and Cox*, 1967]. Samples orientations were determined with both magnetic and sun compasses. Although azimuthal directions were very similar from both methods, the sun compass derived azimuths were used in all cases.

Cores 2.4 cm in diameter were cut to lengths of 2.4 cm for laboratory analysis. Natural remanent magnetizations (NRM) were measured for one specimen from each core using a Molspin Minispin magnetometer; stepwise alternating field demagnetization was done using a single-axis Schonstedt demagnetizer. Thermal studies, using a noninductively wound furnace, were done on selected specimens cut from the same cores that had been stepwise demagnetized using alternating fields.

The Servilleta Basalt (SB) samples from flows exposed in the Rio Grande gorge have NRM intensities of approximately 1 A/m and NRM directions that are well grouped for each site. The samples exhibit stable behavior and directions change only slightly, with improved clustering, after alternating field demagnetization. Median destructive fields are from 40 mT to 80 mT. Typical vector end-point diagrams for cores from both a reversed flow (42TP-326) and a normal flow (50TP-388) show removal of a small overprint and then linear decay to the origin (Figure 3*a*,and 3*b*). Samples from two flows in the southern gorge (49 and 51) show initial large within-site scatter, indicative of lightning strikes. After extensive demagnetization and the use of principal component analysis [*Kirschvink*, 1980] samples from both sites provided reliable data.

The samples from flows exposed on the surface of the TPVF (including Servilleta Basalt, andesite, and dacite) show NRM directions that are scattered within each site. Initial intensities vary greatly, both within a single flow and between flows, with site mean NRM intensities ranging from a few amperes per meter to greater than 100 A/m. Individual samples show large intensity loss and direction change at low levels of demagnetization (Figure 4*a*). In some cases stable directions are reached after demagnetization of 30 or 40 mT (Figure 4*b*), while in other cases stable directions have not been identified after demagnetization to 80 mT. Data for cores from the same site displayed a tendency to

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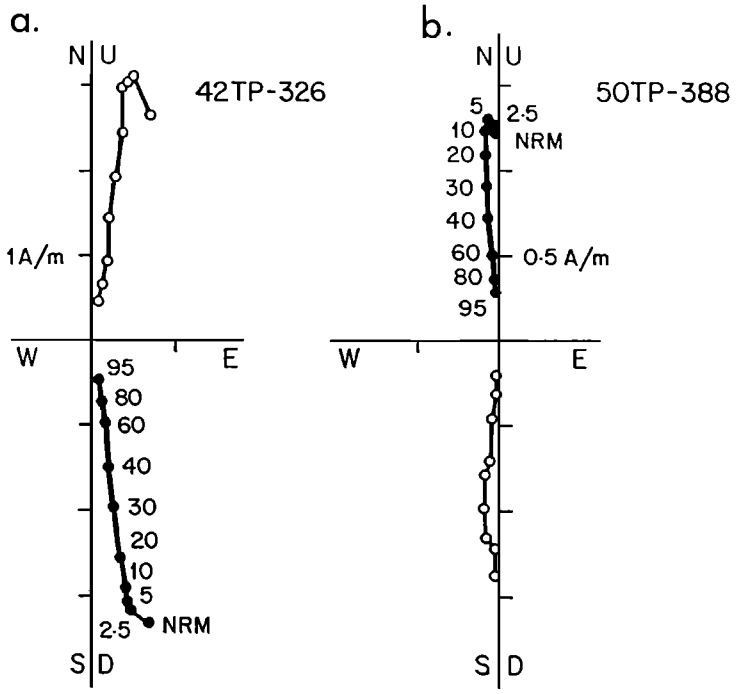


Fig. 3. Vector end-point diagrams for typical flows of the Servilleta Basalt exposed in the Rio Grande gorge, showing both (*a*) reversed and (*b*) normal directions, with simple decay curves and little overprinting. Solid circles are horizontal projections, and open circles are vertical projections; numbers are alternating field demagnetization levels in milliteslas.

converge from widely scattered NRM directions toward a common direction; remagnetization circles were often needed on single samples and on groups of samples from the same flow to identify characteristic directions. Such behavior was seen in both normal and reversed polarity sites and in samples from all the represented rock types. This is similar to the behavior of rocks struck by lightning (a distinct possibility on a high plateau in the desert Southwest), but not all samples exhibited the high intensities usually linked to lightning strikes.

Samples from four SB flows (sites 54-55B) in Garapata Canyon (part of a magnetic excursion sequence discussed later) demonstrate different magnetic behavior from the other SB samples. Intensities are an order of magnitude less than other gorge flows, and NRM directions show greater scatter. Mean coercivities are variable but high (Figure 5*a*), and large direction changes occur during demagnetization (Figure 5*b*). Samples from all four sites yield transitional directions after demagnetization and, in same cases, application of remagnetization circles.

Of the total 62 sites sampled, two sites were excluded from further consideration due to large ($a95 > 20°$) within-site scatter after extensive magnetic cleaning. Thermal demagnetization of selected cores from the entire data set confirms the trends and directions seen on alternating field (AF) demagnetization but does not aid in identifying primary directions in the scattered sites. None of the site data are corrected for regional tilt, given that the Servilleta Basalt shows only a negligible eastward dip of 0.4° [*Lipman and Mehnert*, 1979; *Dungan et al.*, 1984].

Mean magnetic directions of the 60 sites that gave stable directions are plotted in Figure 6, with site data and statistics given in Table 1. Sites 54, 55, 55A, 55B, and 15 have intermediate directions and are not included in the calculations of the mean TPVF direction. The mean directions of the different rock types are all similar at the 95% confidence level, although the small number of sites for all but the Servilleta Basalt result in large confidence limits. Data from reversed sites are inverted to calculate the mean, which is I=49.3°, D=356.7° and $a95$=3.6° (Figure 7). Ten of the 11 flows reported by *Ozima et al.* [1967] from the Dunn Bridge site (in the Rio Grande gorge northwest of Taos) yield a mean direction of I=48.5° and D=354.7°, with $a95$=7.4°. The mean directions from this study are similar to expected directions (I=53.4° and D=357.3°) calculated for this latitude from a mean pole for North America in the late Tertiary, determined by *Brown and Golombek* [1985]. As suggested by Miocene data from the High Lava Plains of Oregon and the Columbia River Basalts [*Mankinen et al.*, 1987], the geocentric axial dipole can be used as a reference field for rocks this young. For the latitude of the Taos Plateau an inclination of 56.4° and declination of 0° is indicated, also in good agreement with the observed data.

REVERSALS AND EXCURSIONS

Correlations of Volcanic Units

Correlations were made between the magnetic directions of Servilleta Basalt sections in the Rio Grande gorge and field and geochemically defined stratigraphic sequences [*Dungan et al.*, 1984]. The southern gorge location of this study (sites 40-51, Figure 2; "Southern Gorge", Figure 8) coincides with "section 4" *of Dungan et al.* [1984]. The Garapata Canyon location (sites 52-59, Figure 2; Figure 8) is identical to their section 10. The Dunn Bridge site [*Ozima et al.*, 1967]; (OZ in Figure 2) is located midway between these two locations. Figure 8 is a compilation of data illustrating Servilleta basalt stratigraphy [*Dungan et al.*, 1984], magnetic polarities (this study); [*Ozima et al.*, 1967], and K-Ar radiometric dates [*Ozima et al.*, 1967; *Mankinen and Dalrymple*, 1979]. An attempt was made to quantitatively correlate magnetic directions in these flows using the technique of *Bogue and Coe* [1981]. Although a few individual flows, which already had identical directions, did correlate when this method was used, the entire sequences proved to be unreliable. This was due not to the fact that the directions are actually different but

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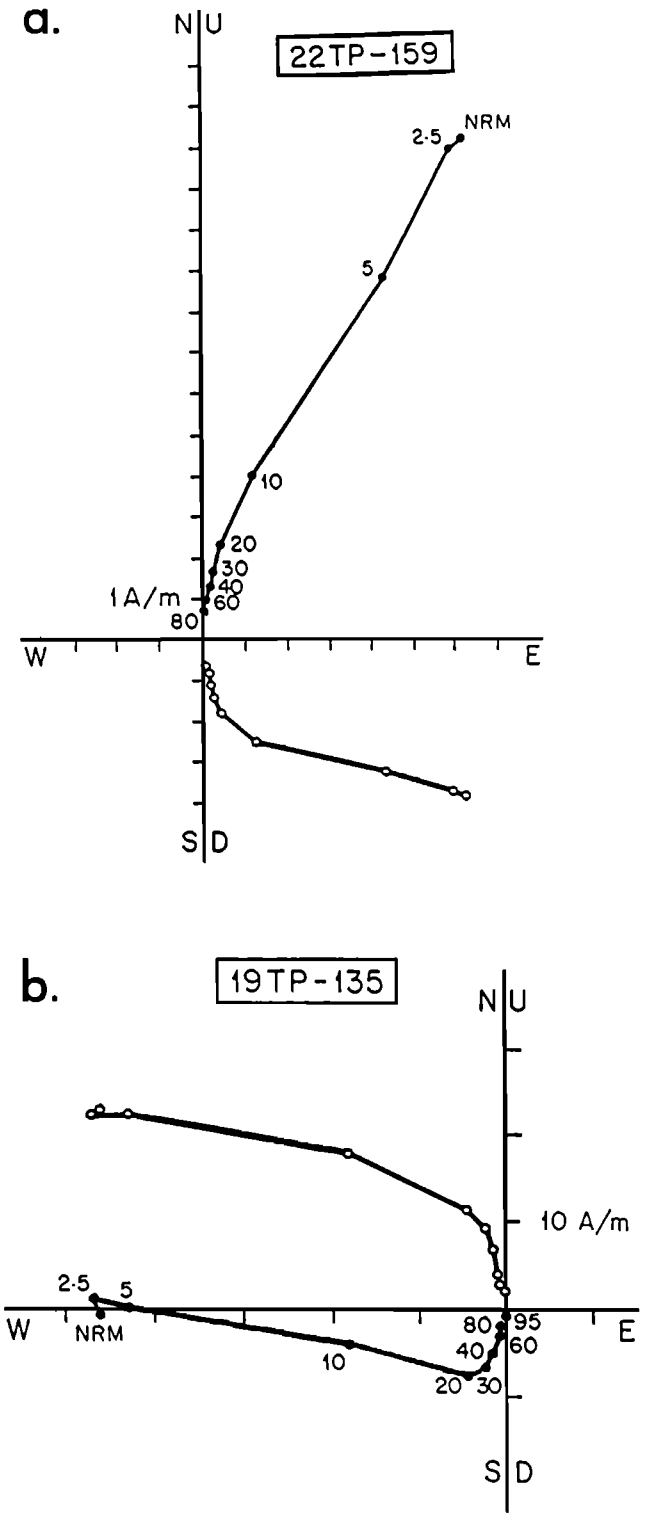


Figure 4. Vector end-point diagrams for units exposed on the surface of the Taos Plateau for normal (a) and reversed (b) polarities, both showing viscous behavior below 30 mT. Symbols as in Figure 3.

rather to the fact that large within-site variances made the statistics inconclusive. As a result, the correlations discussed here are based entirely on polarity.

In the southern gorge section, *Dungan et al.* [1984] have identified the three packages (LSB, MSB, and USB) using geochemistry and stratigraphic locations. All paleomagnetic sites in the lower and middle part of the section as well as the lowermost USB have reversed polarity. The upper part of the USBs, representing the top of the gorge sections and flows covering the nearby plateau, yield normal polarities (Figure 8).

The sequence in Garapata Canyon reveals strikingly dissimilar magnetic directions from the predominantly reversed section at the southern gorge. Using geochemcial data, *Dungan et al.* [1984] identified the lowermost part of the section to be LSB, with a small sequence (three flows) of MSB above. The section then contains a large sedimentary unit, followed by only one USB flow capping the sequence. The lowermost two flows in the section are normal, followed by four flows with transitional directions. The three overlying flows of the MSBs are again all normal polarity, as is the top flow.

The polarity data from Dunn Bridge [*Ozima et al.*, 1967] correlate with the results from the southern gorge section. The same reversed polarity for flows in the lower and middle part of the section is seen, with the only normal polarities observed at the top of the section (Figure 8). There is some uncertainty and confusion concerning the specific sampling locations at Dunn Bridge and their relationship to a fault (down to the west) which intersects the gorge a t Dunn Bridge. This, with the lack of geochemcial data on this section, makes direct correlation between polarities and stratigraphy difficult.

Although our southern gorge data agree with the Dunn Bridge data [*Ozima et al.*, 1967], the LSBs and MSBs with reversed polarities at both these localities do not correlate with the normal and transitional directions at Garapata Canyon. This discrepancy could be due to the fact that lava flows do not represent an uninterrupted, continuous record of the Earth's magnetic field. Alternatively, the flow units in the Taos plateau could be much more restricted in areal extent than previously thought, and these sections sample lava packages separate in space and time.

Potassium-argon dates for samples from the Servilleta Basalt in and near the Rio Grande gorge data of *Ozima et al.* [1967], corrected using the method of *Dalrymple* [1979]; *Mankinen and Dalrymple* [1979]; and *Baldridge et al.*, [1980] range from 4.67 to 3.72 Ma. This suggests an emplacement time of roughly 1.0 m.y for the entire central Rio Grande gorge sequence. The dates fall within the Gilbert Reversed-Polarity Chron, a period of predominantly reversed polarity with numerous subchrons of normal polarity [*Mankinen and Dalrymple*, 1979]. *Dungan et al.* [1984] suggest on the basis of geochemical similarities and thicknesses of interbedded sediments, that the flow packages were deposited in spurts of several hundred years duration with much longer periods of inactivity between flows. Therefore large portions of the magnetic polarity time scale may not have been recorded by the Servilleta Basalt, and although some flows correlate chemically, they may represent different polarity intervals.

Garapata Canyon Excursion

The transitional directions of the four LSB flows in Garapata Canyon describe an excursion of the magnetic field. Sites 52 and 56, below and above the transitional sites, are both normal and behave as typical SBs with little change in direction during demagnetization. The transitional flows have lower intensities (by an order of magnitude) and relatively complex demagnetization curves (Figure 5*b*). The high median destructive fields (Figure 5*a*), linear decay of the demagnetization vector end-points, and within-site stability indicate that the component isolated above 20 mT demagnetization is a stable primary direction.

The inclinations and declinations for the Garapata Canyon sites (flows 52-58, Figure 8) are listed in Table 1. The lowermost flows (and lowest flows exposed in this area) have normal polari-

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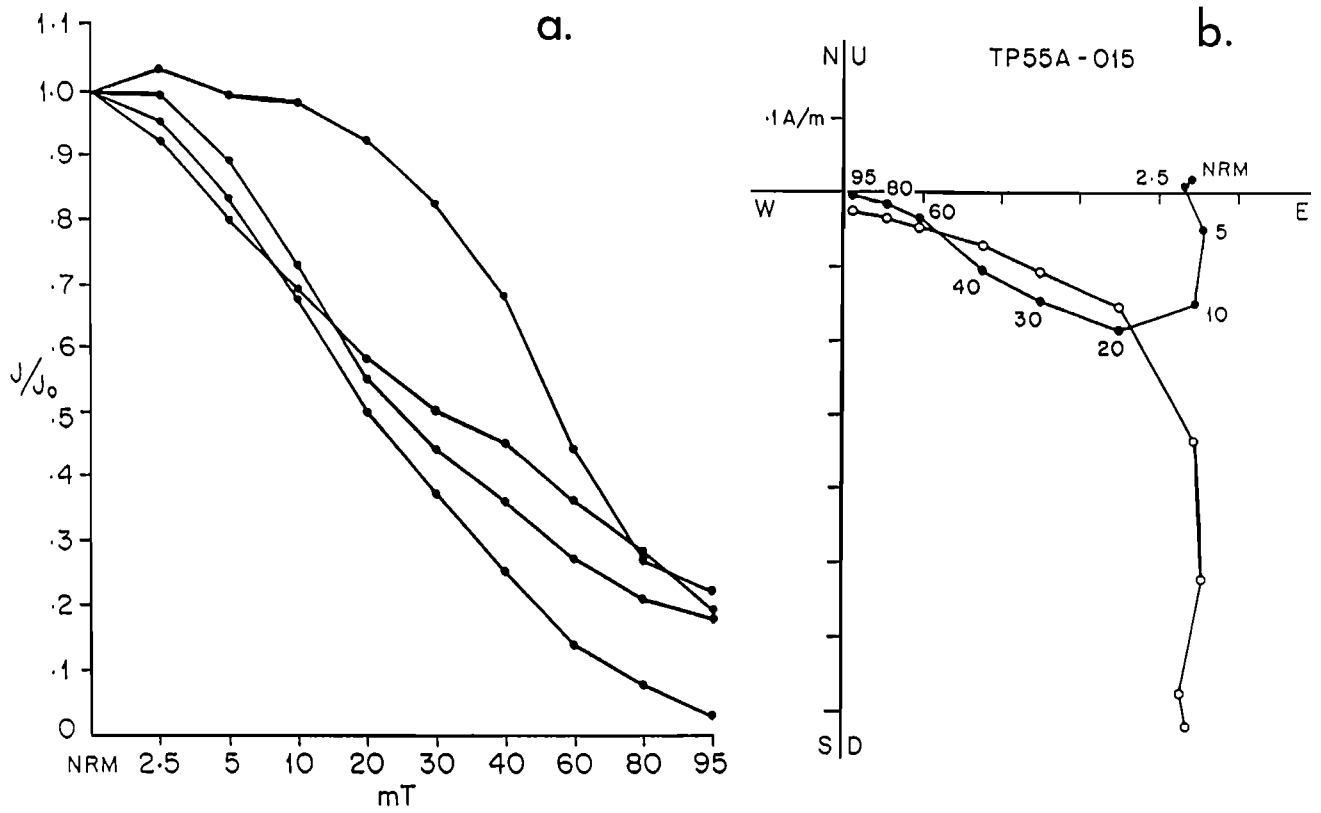


Fig. 5. Demagnetization data on flows of the Garapata Canyon excursion: (*a*) normalized intensity plot for four samples, one from each of the four excursion flows and (*b*) typical vector end-point diagram for sample from the excursion. Symbols are as in Figure 3. Data show high mean destructive fields and considerable direction change during demagnetization.

ties. Continuing along a great circle path and stratigraphically up section, there is an increase in declination, a shallowing of inclination, and a decrease in intensity. As the excursion proceeds, there is a return to normal polarity along a great circle path at steeper inclinations and an increase in intensities to the original levels. Flows 57 and 58 are separated from the transition flows by a thin soil horizon, but they lie directly along the great circle path of the return to normal polarity and are therefore considered to be part of the excursion. This can be seen in Figure 9*a* where the Garapata Canyon section data are projected down the mean axial dipole field direction for the site latitude [e.g., *Hoffman*, 1984]. The data describe a well-defined excursion with a slight degree of near-sidedness. Figure 9*b*, a global projection of the virtual geomagnetic poles (VGPs) of the Garapata excursion, shows that the largest departure of VGPs from geographic north is reached at latitude 22°S and longitude 320°E, on the east coast of South America. It is interesting to note that this location conicides with one of two global "cluster patches" of transitional directions identified by *Hoffman* [1992]. Data from both complete reversals and excursions show a preference for these positions, indicating the possible influence of the lower mantle on the core dynamo [*Hoffman*, 1992]. General correlation to the dates from Dunn Bridge [*Ozima et al.*, 1967] places the excursion observed here in the Gilbert Chron, but further refinement of age is not possible with the available data.

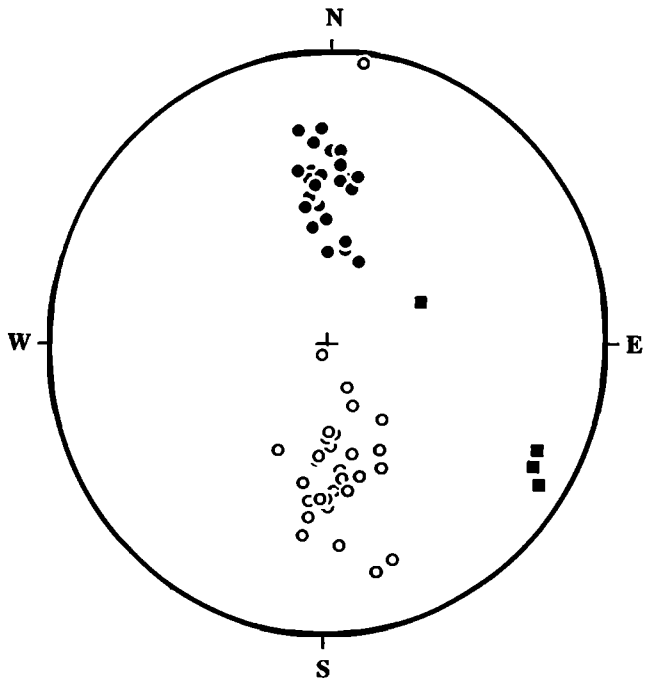


Figure 6, Stereographic projection of site mean data after demagnetization for TPVF. Open symbols are projections on the upper hemisphere, closed symbols are projections on the lower hemisphere. Circles, assorted units; squares, Servilleta basalts recording excursion.

Inclination Anomaly and Paleosecular Variation

Using the mean inclination (49.3°) for the TPVF, an inclination anomaly (dI) of -6.7° is obtained between the observed inclination and the inclination expected from the geocentric axial dipole. Late Quaternary studies at similar latitude from California and Tennessee yield dI values of -5.5° and -3.9°, respectively [*Lund*, 1985]. These data points agree at the 95% level with worldwide averaged results for the past 5 m.y. by *Lee* [1983], as reported by *Merrill and McElhinny* [1983]. This analysis distinguished

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TABLE 1. Magnetic Direction, Taos Plateau Volcanic Field

Site	Lat	Long	N	J_{NRM}	I	D	a_{95}	k	PLA	PLO
					Servilleta Basalt					
2	36.78	254.02	2/3/7	1.2	-86.4	199.7	9.1	92	-43.5	77.3
3	36.78	254.02	6/6	15.6	-63.5	178.9	5.9	93	-81.6	67.2
4	36.74	254.02	3/6/10	102.6	42.6	7.6	7.9	49	76.2	43.7
5	36.65	254.04	7/7	21.1	34.6	0.5	6.9	58	72.4	72.5
6	36.65	254.04	7/7	78.9	27.1	351.6	12.8	17	66.5	94.8
7	36.34	253.94	7/7	5.7	-34.3	186.6	7.2	54	-71.6	233.8
8	36.78	254.02	4/3/7	40.0	-64.1	173.2	8.4	58	-79.6	46.8
10	36.82	254.02	3/5/8	69.0	-65.3	177.2	8.2	53	-79.2	63.8
13	36.92	254.01	6/7	18.2	50.6	355.8	6.8	97	83.4	107.1
20	36.76	254.27	2/4/9	26.6	34.1	3.7	8.6	78	71.7	63.1
21	36.35	254.26	8/8	3.4	40.1	354.1	2.3	591	75.6	96.6
22	36.36	254.20	7/7	9.6	38.4	3.6	5.4	127	74.9	61.2
23	36.36	254.15	5/3/8	11.0	-62.8	142.2	10.6	35	-60.4	11.4
24	36.44	254.09	2/4/7	23.3	-31.2	175.6	8.4	81	-70.0	266.5
25	36.53	254.05	8/1/9	13.9	57.0	351.8	6.3	69	83.4	175.7
26	36.61	253.98	5/1/9	43.0	48.6	353.7	12.4	31	82.1	112.6
27	36.75	254.33	6/7	11.4	42.7	352.9	5.6	142	76.6	103.2
29	36.72	254.16	4/3/8	69.5	-20.5	166.5	9.2	64	-61.2	282.6
30	36.62	254.00	2/5/9	61.1	48.2	352.1	7.6	77	80.1	118.2
32	36.57	254.21	7/8	12.3	-44.8	186.1	5.6	119	-78.6	225.5
34	36.51	254.24	9/11	7.6	31.7	355.6	5.6	85	70.3	86.8
35	36.48	254.26	4/2/8	33.4	27.0	358.1	8.8	64	67.7	79.1
40	36.44	254.27	7/7	2.2	-51.0	164.7	3.5	299	-76.5	328.2
41	36.44	254.27	6/6	3.4	-51.1	172.1	3.4	390	-82.0	311.1
42	36.44	254.27	6/7	5.9	-47.2	170.6	2.8	579	-78.7	301.4
43	36.44	254.27	8/8	4.9	-39.8	185.1	7.4	57	-75.5	235.2
44	36.44	254.27	7/7	1.5	-49.7	189.3	3.1	372	-80.3	199.0
45	36.44	254.27	7/7	2.2	-43.0	178.7	3.8	257	-78.5	260.2
46	36.44	254.27	7/7	1.8	-45.7	179.0	4.4	192	-80.7	259.8
47	36.44	254.27	7/7	2.5	-45.7	181.8	3.3	328	-80.6	244.5
48	36.44	254.27	8/8	2.2	-50.8	154.6	2.0	778	-68.4	338.5
49	36.44	254.27	8/8	20.5	-55.9	151.9	(3.3,4.7	-167,-2)	-66.8	353.0
50	36.44	254.27	8/8	2.2	41.7	357.0	3.6	240	77.3	86.8
51	36.44	254.27	2/4/7	45.1	31.7	355.3	9.2	68	70.3	87.7
52	36.61	254.31	7/7	1.4	43.7	4.4	2.0	937	78.3	54.3
53	36.61	254.31	7/7	3.5	45.7	9.1	2.7	519	77.8	32.6
54*	36.61	254.31	6/7	0.8	16.9	115.8	4.6	217	-14.8	321.3
55*	36.61	254.31	7/7	0.8	10.5	122.5	7.1	73	-22.0	319.2
55A*	36.61	254.31	4/4/8	0.9	15.5	119.7	5.8	102	-18.2	319.2
55B*	36.61	254.31	5/1/7	2.5	60.4	67.6	11.9	34	38.6	316.9
56	36.61	254.31	8/8	3.2	64.3	359.6	1.9	859	80.5	252.6
57	36.61	254.31	6/6	3.4	63.1	11.3	4.4	237	78.3	297.8
58	36.61	254.31	7/7	3.5	65.4	21.5	3.3	341	70.8	303.0
59	36.61	254.31	12/14	22.9	39.5	349.5	4.6	89	73.1	110.3
60	36.28	253.94	5/2/9	21.0	-56.7	204.2	10.9	33	-70.6	153.7
Mean			41/45		47.4	357.2	4.2	230	81.7	91.6
					Xenocrystic Basaltic Andesite					
11	36.88	254.06	7/8	38.6	-57.1	183.8	9.3	43	-86.9	147.7
12	36.89	254.06	3/4/7	16.9	-53.6	173.0	11.3	33	-83.7	320.4
16	36.83	253.94	1/5/8	109.8	50.8	349.5	6.7	143	79.8	135.5
17	36.82	253.98	6/8	9.6	41.6	10.4	(8.3,10.2	-42,-1)	74.3	36.3
18	36.79	254.01	3/5/8	31.4	-70.6	156.0	8.8	46	-65.6	39.4
19	36.78	254.06	4/2/6	18.7	-57.8	183.1	11.8	36	-87.0	128.9
Mean			6/6		55.7	357.9	9.6	50	88.3	152.8
					Andesite					
14	36.90	254.02	6/2/8	47.2	44.6	354.8	5.0	129	78.5	98.0
15*	36.89	254.03	3/5/9	169.2	-3.9	7.2	10.9	30	-50.6	242.7
33	36.55	254.24	8/8	23.0	-76.2	152.5	7.8	51	-58.1	51.6
36	36.69	254.38	8/8	12.7	55.0	359.2	11.8	23	88.7	103.7
Mean			3/4		58.9	353.0	26.5	23	83.7	195.3

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TABLE 1. (continued)

Site	Lat	Long	N	J_{NRM}	I	D	a_{95}	k	PLA	PLO	
					Dacite						
28	36.72	254.33	7/8	37.3	61.3	10.4	(6.3,10.0	-51,-2)	80.3	295.4	
31	36.59	254.16	8/8	4.7	-22.7	161.9	10.3	30	-60.3	292.1	
37	36.58	254.83	8/8	1.8	-57.5	166.2	5.2	115	-78.9	356.5	
38	36.57	254.82	7/8	2.4	-47.6	175.8	2.9	428	-81.4	279.7	
39	36.54	254.83	9/10	1.2	-60.7	176.4	3.0	302	84.1	227.5	
MEAN			5/5		50.5	352.3	16.9	22	81.7	126.7	
TPVFMEAN			55/60		49.3	356.7	3.6	29	83.1	98.9	
EXPECTED†						53.4	357.3	3.5		86.5	114.7

Lat and Long are north latitude and east longitude, in degrees, of each site location. N is number of samples (sites) used in calculations: number of samples (sites) measured or, number from direct observations / number from remagnetization circles / total number measured. J_{NRM}, is mean intensity of all samples measured for natural remanent magnetizations in amperes per meter. *I* and *D* are mean inclination, positive downward, and declination, east of north, in degrees. a_{95} is radius of cone of 95% confidence around mean direction; *k* is precision parameter [*Fisher*, 1953], precision parameter for mixed sites from *McFadden and McElhinny* [1988]. Numbers in parentheses are parameters for Bingham statistics [*Onstott*, 1980], determined using remagnetization circles. PLA and PLO are pole latitude, positive northward, and pole longitude, eastward, in degrees.

Transitional directions are not included in mean calculations.

† Expected direction from reference pole of *Brown and Golombek* [1985], calculated for latitude of Taos Plateau volcanic field (TPVF).

between normal and reversed polarity information; if only normal sites from the TPVF are used, a dI of -10.7° is obtained, which is statistically significant from the global data. Only a very small part of this low value is possibly due to tectonic tilting in the southern part of the plateau (see discussion below), because normal sites predominate in the northern part of the sampling area, where little or no shallowing has occurred.

Angular dispersion of the VGPs of the TPVF, excluding the five excursion sites, was calculated using standard methods. As reported in Table 2, the angular standard deviation for the TPVF, calculated with respect to the geographic axis, is 16.3°, with upper and lower 95% errors of 18.9° and 14.4°, respectively [*Cox*, 1969]. This observed VGP scatter is similar to that predicted by the most recent model comparing secular variation to latitude. Model G [*McFadden et al.*, 1988] provides a variation of VGP scatter with respect to latitude dependent on the contributions of the dipole and quadrupole families. This curve is shown in Figure 10 fit to averaged global data for the past 5 m.y. [*McFadden et al.*, 1988] along with the VGP dispersion for the Taos data. Other published angular standard deviations for areas in North America at similar latitudes are given in Table 2. These include anomalously low paleosecular variation recorded for the western United States [*Doell and Cox*, 1972]; (WUS1, Table 2) and Anderson Pond, Tennessee [*Lund*, 1985]; (APT, Table 2), but both these studies deal only with late Quaternary data. Data on lava flows in eastern Arizona younger than 2.1 Ma yield a dispersion of 18.0° [*Castro et al.*, 1985]; (AZ, Table 2), greater than expected from the model. A compilation of western United States data spanning the last 5 m.y. [*McElhinny and Merrill*, 1975]; (WUS2, Table 2) yields an angular dispersion in close agreement with that predicted by model G.

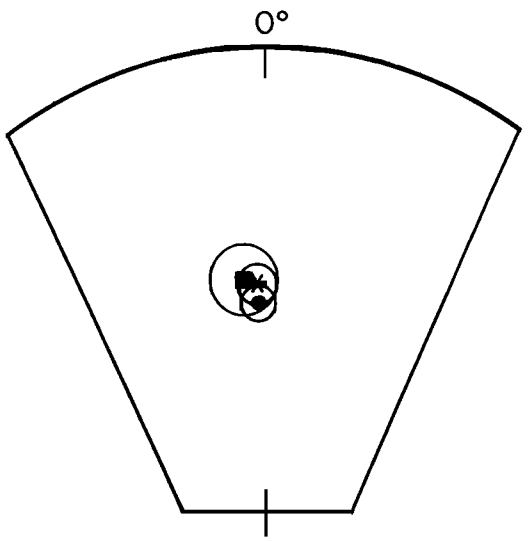


Figure 7. Portion of stereographic projection comparing the mean direction for the TPVF (asterisk) to the expected direction at this latitude (dot). Tick marks are 20°. Square is mean for Dunn Bridge location of Ozima et al.(1967). All means shown with circles of confidence at the 95% level.

The observations that dI is low but that paleosecular variation for the TPVF is as expected are not inconsistent. As noted by *Lund* [1985], these two phenomena do not have to be generated by the same processes and can be considered independent. It is also well known that "spot" readings of the magnetic field, as represented by dI or paleosecular variation, may be spurious [*Merrill and McElhinny*, 1983]. However, consistent data over a regional area and averaged over millions of years suggests that variations of the nondipole field may be much more complex than presently supposed [*Brown and Castro*, 1987].

TECTONIC ROTATIONS

In an effort to investigate the tectonic behavior of the southern San Luis basin during the last 5 m.y., mean inclination and declination for the TPVF were each compared to the expected inclination and declination. Changes in declination from the expected declination can be viewed as rotations (horizontal movements around a vertical axis). Flattenings, changes in the inclination compared to the expected inclination, represent a north-south translation [*Beck*, 1980], or tilt around a horizontal axis.

Using a North American reference pole for the late Tertiary determined by *Brown and Golombek* [1985] and the methods of *Beck* [1980] and *Demarest* [1983], the rotation and flattening

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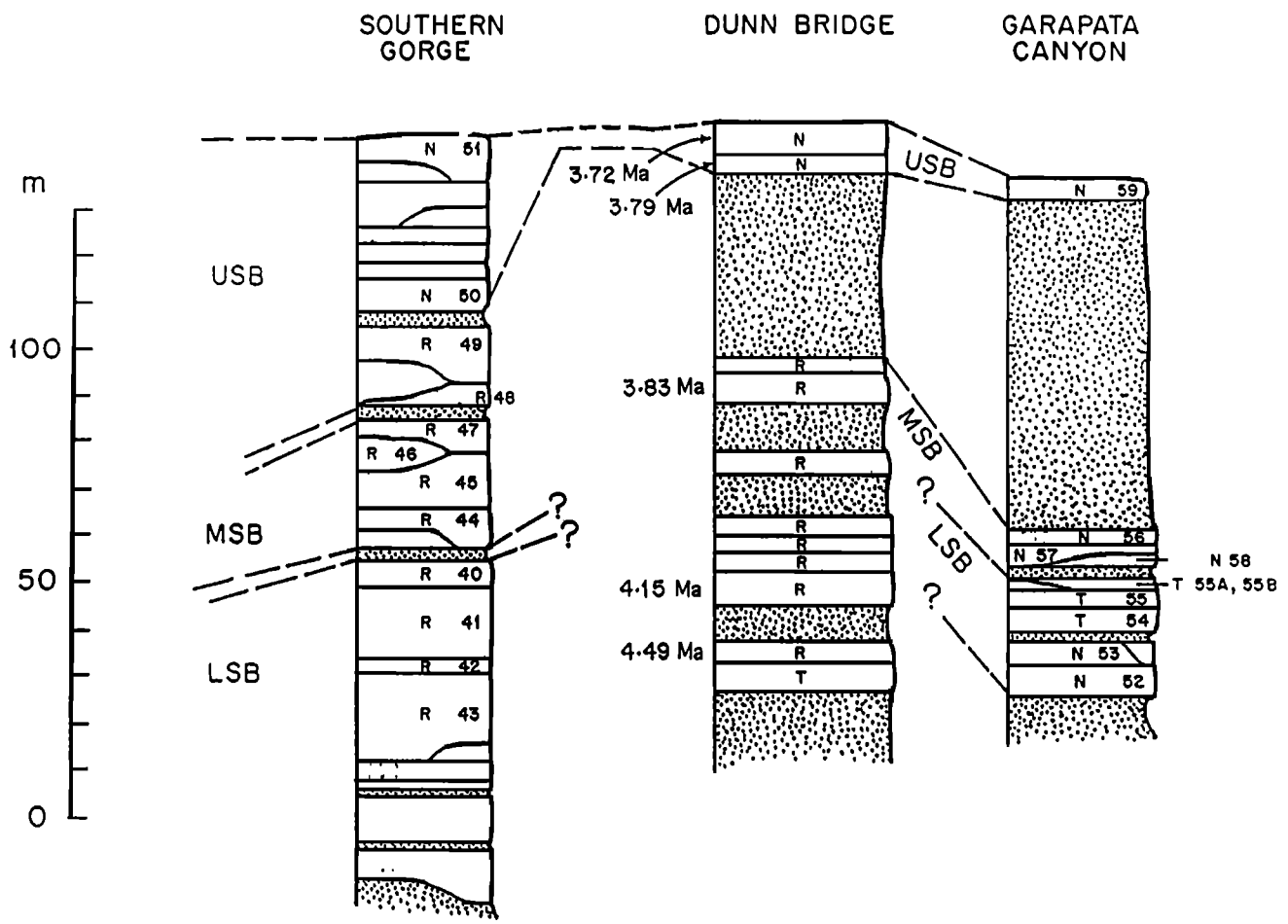


Fig. 8. Stratigraphic and magnetostratigraphic sections of Servilleta basalts in the Rio Grande gorge. Patterns: white, lava flows; stippled, sediments and/or rubble zones. N is normal polarity; R is reversed polarity. USB, MSB, LSB are upper, middle, and lower Servilleta Basalt; numbers refer to paleomagnetic sites listed in Table 1. Southern Gorge and Garapata stratigraphy and Servilleta Basalt correlations are from *Dungan et al.* [1984]; Dunn Bridge stratigraphy and polarity are from *Ozima et al.* [1967]. Radiometric dates are from *Ozima et al.* [1967], corrected using new decay constants [*Dalrymple*, 1979].

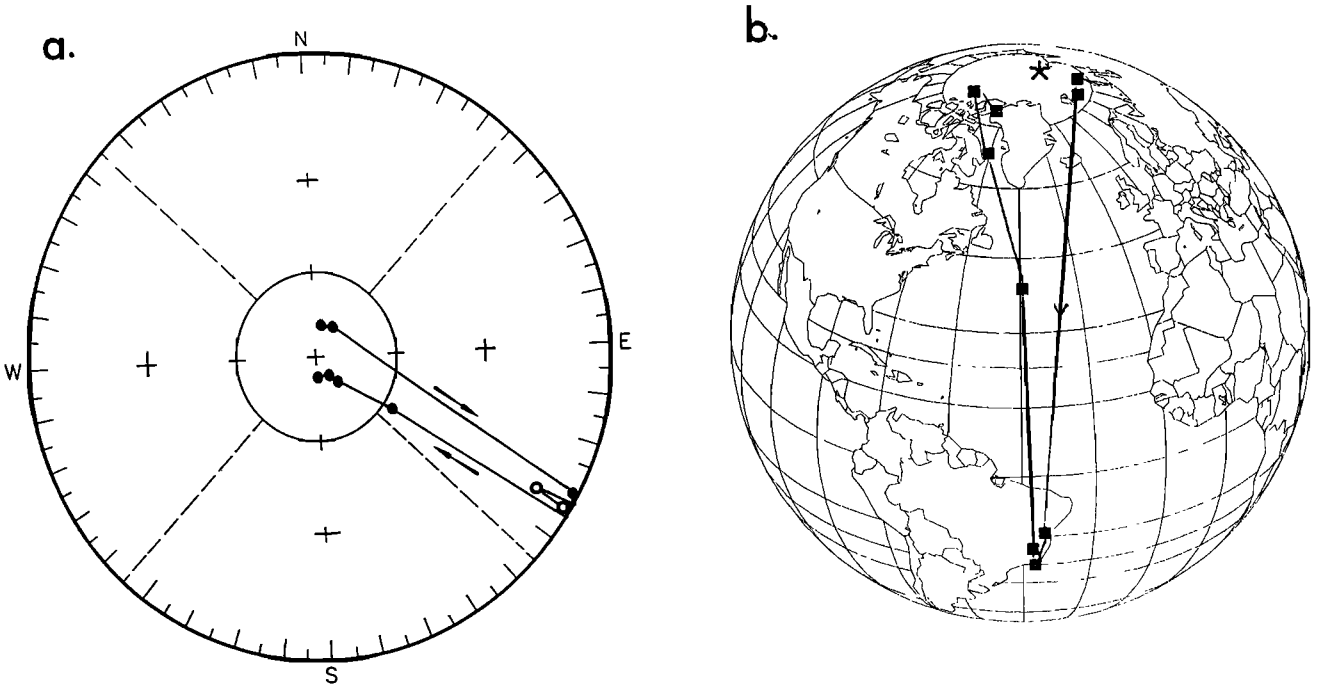


Fig. 9. Stereographic projections of Garapata Canyon excursion. (*a*) Plot of inclination and declination, centered on axial dipole direction for site [*Hoffman*, 1984]; open symbols indicate upper hemisphere and solid symbols indicate lower hemisphere. (*b*) Plot of virtual geomagnetic pole date for excursion; star is mean pole for Taos Plateau volcanic field data.

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TABLE 2. Paleosecular Variation, Western United States

Site	Lat	N	S_f	S_u	S_l	Age	Reference
AZ	34.0	46	18.0	21.1	15.7	<2.1	*Castro et al.* [1985]
TPVF	36.5	52	16.3	18.9	14.4	1.8 - 4.7	This study
WUS2	37.0	76	16.3	18.4	14.7	<5.0	*McElhinny and Merrill* [1975]
APT	37.0	-	11.6	12.8	10.6	<0.02	*Lund* [1985]
WUS1	38.0	21	12.5	15.9	10.3	<0.7	*Doell and Cox* [1972]
SV	38.5	25	18.5	23.1	15.4	3 - 8	*Mankinen* [1989]

Lat is degrees north latitude; N is number of sites; Sf is angular dispersion of VGP data, in degrees; S_u and S_l are upper and lower 95% error limits [*Cox*, 1969]; age is in millions of years. Sites are AZ, Springerville Volcanic Field, Arizona, TPVF, Taos Plateau Volcanic Field, WUS2, Western United States, APT, Anderson Pond, Tennessee, WUS1, Western United States, SV, Sonoma Volcanics, California.

have been calculated (Table 3). The 55 nonexcursion sites of the TPVF yield a rotation of -0.6° ± 5.6° and a flattening of 4.1° ± 4.1°. These data indicate no rotation of the TPVF; although to the south, the Espanola and Jemez blocks may have been rotated approximately 15° counterclockwise and clockwise, respectively [*Brown and Golombek*, 1985; 1986].

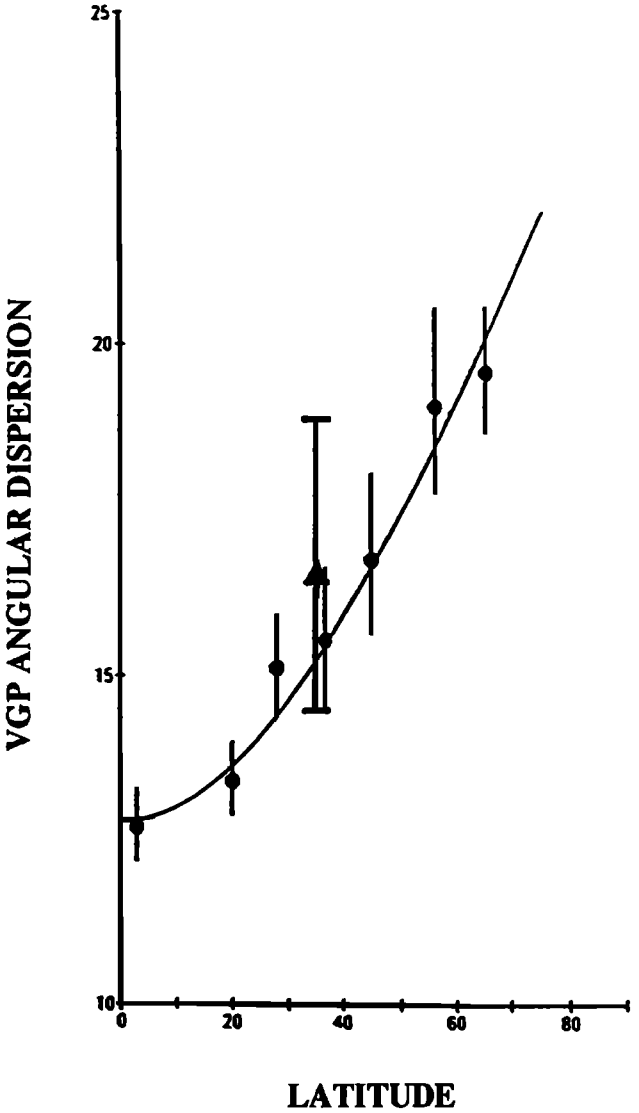


Figure 10. Angular dispersion of virtual geomagnetic poles plotted with respect to latitude. Dispersion of poles from the TPVF (triangle) with 95% confidence limits plotted on secular variation model G of McFadden et al. (1988). Solid line is model G fit to averaged global data 0-5.0 Ma (solid dots).

Considering all of the TPVF data, there is a small amount of flattening (4.1° ± 4.1°), but it is not statistically significant. To test whether flattening is observable in localized regions, the data were subdivided into geographic areas (north, mid, and south TPVF; east and west TPVF). This grouping was also done to test the hypothesis of bowing of the TPVF or downwarping of the southern plateau as proposed by *Muehlberger* [1979] and *Dungan et al.* [1984] due to recent NW-SE compression from the overriding of the Picuris Mountains at the plateau's southeast corner. If there has been recent (within the last 5 m.y.) arching of the plateau, the geometry of the magnetic directions, in relation to the tilting, would require that the inclinations at sites on the northwest limb of an arch with a NE-trending fold axis would steepen and that the inclinations on the southeast limb would become more shallow (e.g., Figure 11). Although the error bars on the flattening of the northern and mid TPVF would indicate these are not significant, there is a definite trend from north to south (Table 3). The northern group has steepened slightly (-0.4° ± 6.2°), the mid TPVF group reflects the mean flattening (4.2° ± 6.2°), and the southern group is shallower than the mean, with a statistically significant flattening of 8.3° ± 5.3°. A similar pattern of tilting is seen in an east-west comparison, with the eastern half of the area indicating some downwarping (flattening of 5.7° ± 4.6°) and the western half indicating negligible tilt. These data support the suggestion that the TPVF has acted as a stable buttress against which the Espanola block to its south has rotated. As a result, the Picuris Mountains, which comprise the Espanola block's northeast corner, have been thrust up and over the TPVF, thereby downwarping the plateau at its southern end. It appears that for this compression to occur, the rate of extension across the southern Taos graben, the recently active sub-basin of the San Luis basin, must be less that the rate of rotation or westward translation of the northern corner of the Espanola block. If the rate of rotation or western translation of the northern Espanola block were less than the extension in the southern San Luis basin near the Embudo fault, no compression and uplift would occur. Evidence suggests some compression across the southwestern Embudo fault [*Aldrich*, 1986], which could be due to convergence from the clockwise rotation (Figure 11) of the Jemez block [*Brown and Golombek*, 1985]; it is not clear if the southwestern corner of the Taos Plateau and/or Tusas Mountains have been similarly compressed.

The warping and lack of rotation of the TPVF show that it has acted as a stable buttress to the rotation of the Espanola block to the south. The reason for rotation of crustal blocks to the south, but not in the San Luis basin is not known, although we can speculate on contributing factors. The long axis of the diamond-shaped Espanola block strikes NE and its bounding faults strike NE and NNE, which are at a slight angle from perpendicular to the least principal horizontal stress (E-W) that has existed

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Table 3, Flattenings and Rotations, Taos Plateau Volcanic Field

Location	N	I	D	a_{95}	F	dF	R	dR
TPVF Mean	55	49.3	356.7	3.6	4.1	4.1	-0.6	5.6
NTPVF	18	53.8	357.9	6.8	-0.4	6.2	0.6	9.8
MTPVF	18	49.2	356.5	6.8	4.2	6.2	-0.3	9.0
STPVF	19	45.1	355.9	5.5	8.3	5.3	-1.4	7.2
ETPVF	32	47.7	356.9	4.4	5.7	4.6	-0.4	6.2
WTPVF	23	51.5	356.4	6.3	1.9	5.8	-0.9	8.8
Expected		53.4	357.3	3.5				

NTPVF, MTPVF, STPVF, ETPVF, and WTPVF are collections of sites in this study subdivided into north, middle, and south, and again into east and west on geographic location; *F* and *R* are flattening and rotation with respect to the expected direction; *dF* and *dR* are errors corresponding to F and R calculated using equations given by *Brown and Golombek* [1985] with error corrections given by *Demarest* [1983].

throughout the region for at least the past 5 m.y. [*Zoback and Zoback*, 1989; *Lipman*, 1981; *Zoback et al.*, 1981; *Golombek et al.*, 1983; *Aldrich et al.*, 1986]. This oblique geometry could result in a component of left slip along the Espanola block that could drive its observed counterclockwise rotation. In contrast, the recently active portion of the southern San Luis basin (Taos graben) is bounded by faults that strike more nearly due north, perpendicular to the least principal horizontal stress (no component of strike slip), and so does not rotate. In addition, if Kelley's model of north-shallowing basins that step to the right along en echelon relay faults is correct, this geometry could also contribute to the left slip [e.g., *Kelley*, 1979; 1982]. Finally, it is interesting to note that the clockwise rotated Jemez block (adjacent to the Colorado Plateau) lies in a different least principal horizontal stress field that is oriented predominantly NW-SE [*Aldrich et al.*, 1986], which could result in the clockwise rotation as originally suggested by *Golombek et al.* [1983] and elaborated upon by *Brown and Golombek* [1985].

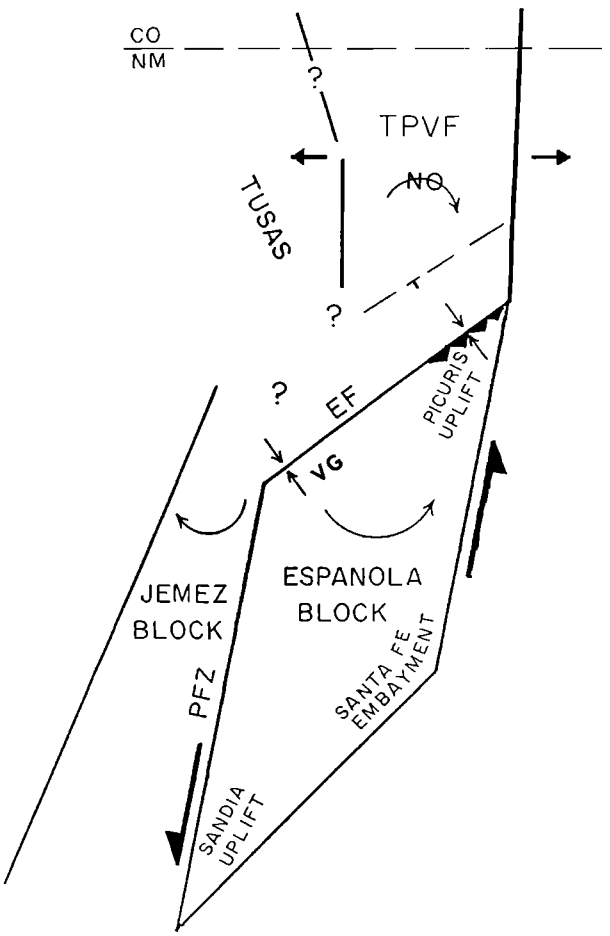


Figure 11. Schematic diagram of rotated and unrotated blocks in the Rio Grande rift, northern New Mexico. EF, Embudo fault; PFZ, Pajarito fault zone, VG, Velarde Graben. The Jemez block has rotated clockwise, the Espanola block has rotated counterclockwise, and the TPVF shows no rotation in the past 5 m.y. Counter-clockwise rotation of the Espanola block has caused convergence at its NE corner (Picuris Uplift). This convergence has resulted in NW-SE compression and thrust motion along the end of the Embudo fault, uplift and overriding of the Taos Plateau by the Picuris Mountains, and downwarping of the southern Taos Plateau.

SUMMARY AND CONCLUSIONS

1. A total of 533 samples were collected from 62 sites in the Servilleta Basalt, basaltic andesite, andesite, dacite, and rhyolite of the Taos Plateau volcanic field. Samples from 60 sites yielded stable directions, and 55 were used to calculate a mean direction for the TPVF of I=49.3° and D=356.7° ($a95$=3.6°).

2. Demagnetization behavior of samples from sites in the Rio Grande gorge and normal polarity sites on the surface of the plateau is straightforward, with simple linear decay to the origin. Reversed polarity samples from sites on the plateau surface show some overprinting and considerably more scatter after demagnetization than do the normal sites. Samples from sites representing an excursion of the magnetic field show low intensities and more complex demagnetization paths.

3. The magnetic directions of 4.7 to 3.6 Ma Servilleta basalts in the Rio Grande gorge reflect normal, intermediate, and reversed polarities within the Gilbert Chron. The magnetostratigraphy of flows in the southern gorge, Dunn Bridge, and Garapata Canyon localities do not correlate, even though their geochemistry indicates similar packages of flows. One explanation for this is that the flows were implaced in short time periods (several hundreds of years) with longer intervals between eruptive events, and thus the flows reflect different magnetic polarity time periods.

4. Data for nine basalt flows in Garapata Canyon reflect an excursion of the magnetic field during a period of normal polarity. The fullest extent of the excursion is marked by a virtual geomagnetic pole at 22°S latitude and shows a slight degree of near-sidedness.

5. The inclination anomaly for the TPVF is -6.7°, agreeing with global data averaged over the last 5.0 m.y. The angular dispersion of virtual geomagnetic poles from the TPVF is 16.3°, similar to expected values from the current model for paleosecular variation.

6. Although rocks to the immediate south have been rotated both clockwise (Jemez block) and counterclockwise (Espanola block), the paleomagnetic data for the TPVF indicate no sign of rotation. (R = -0.6° ± 5.6°). It is suggested that the TPVF has acted as a stable buttress to the crustal blocks to the south.

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7. The flattening of the magnetic inclinations in the TPVF is statistically insignificant (4.1° ± 4.1°), but separation of the data reveals significant flattening in the southern (8.3° ± 5.3°) and eastern (5.7° ± 4.6°) parts of the plateau. This finding agrees with downwarping and bowing of the TPVF due to NW-SE compressional stresses created by the overriding of the Picuris Mountains to the southeast. It also supports the concept of the southern San Luis basin and TPVF as a stable buttress and requires that the rate of extension across the southern Taos graben and San Luis basin be less than the rotation and westward translation of the northern corner of the Espanola block.

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